

# Response to the Editor and Reviewers

**Submission: HITS / BMC Cardiovascular Disorders**

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**Draft limitation:** The exact editor letter, Reviewer 1 and Reviewer 2 reports, and portal-integrated manuscript snapshot were not recovered. The items below are explicitly non-verbatim summaries from the adopted audit instructions and are not quotations. This draft requires comparison with the original reports before submission.

We thank the Editor and reviewers for their careful assessment. We revised the manuscript within the scope of the approved analyses and have not added new model searches, biomarkers, phenotypes, datasets, or algorithms.

## E1

**Comment summary (non-verbatim):** CBC/fibrinogen selection and measurement timing

**Response:** We re-audited the source export and processing rule. A patient-level row was selected using discharge-diagnosis availability, completeness of absolute neutrophil, lymphocyte, monocyte and platelet values, a parsable CBC timestamp, overall completeness, and source-row order as the final tie-break. This was not an earliest- or admission-first rule. The extract did not permit uniform linkage to the index angiography hospitalization, admission, diagnosis, or treatment time; we therefore removed any claim of admission-first, pre-angiography, or pre-diagnostic sampling and state that all 1,820 selected measurements remain timing-unverified. Fibrinogen timing is likewise not assumed.

**Changes in the revised files:** Revised manuscript pp. 4, 5, 8, 9, 18; Additional file 1 pp. 6; Methods: Predictors and laboratory measurements; Results: Study population; Figure 1; Limitations.

## E2

**Comment summary (non-verbatim):** AMI label provenance and clinical adjudication

**Response:** The AMI phenotype was derived from discharge-diagnosis text using explicit acute/subacute MI and subtype wording. We now state that no independent cardiologist, ECG, imaging, or uniform biomarker adjudication was performed for all patients and do not describe the phenotype as formal Fourth Universal Definition adjudication.

**Changes in the revised files:** Revised manuscript pp. 4; Additional file 1 pp. 1, 6; Methods: Outcome definition; Discussion: limitations; Additional file 1; Table S2.

## E3

**Comment summary (non-verbatim):** Reconciliation of source records and unique patients

**Response:** The flow now distinguishes 2,548 source records from 2,279 unique patients. Of these, 2,010 had one source record and 269 had two; we describe these as multiple source records, not proven repeat admissions. One row per patient was retained by the stated completeness-based rule.

**Changes in the revised files:** Revised manuscript pp. 4, 9; Methods: Eligibility and study population; Results: Study population; Figure 1.

## E4

**Comment summary (non-verbatim):** Nested/repeated validation, aggregation, and bootstrap intervals

**Response:** We explicitly report stratified outer five-fold cross-validation repeated 10 times, five-fold inner tuning, one held-out probability per patient per repeat, arithmetic averaging of the 10 held-out probabilities, matched outer partitions for paired comparisons, and training-fold-only preprocessing and tuning. The 1,000 bootstrap intervals resample fixed mean out-of-fold predictions and outcomes without refitting; we now state that they condition on those predictions and omit full model-development uncertainty.

**Changes in the revised files:** Revised manuscript pp. 6; Additional file 1 pp. 2; Methods: Internal validation and uncertainty; Additional file 1; Additional file 2, Table S6.

## E5

**Comment summary (non-verbatim):** Date-axis analysis and temporal generalizability

**Response:** We renamed this analysis the exploratory deidentified date-axis sensitivity analysis. The prespecified groups were development 1001/196 AMI, buffer 271/79 AMI, and later 548/178 AMI. Later-group Core-7 and PIV AUCs were 0.701 and 0.699; paired  $\Delta$ AUC was 0.002 (95% CI -0.021-0.025). The advantage attenuated. Because dates could not be uniformly linked to the index angiography hospitalization, we do not call this formal temporal validation or claim transportability.

**Changes in the revised files:** Revised manuscript pp. 7, 14; Additional file 1 pp. 4; Methods: Temporal sensitivity analysis; Results; Figure 4; Discussion: limitations.

## E6

**Comment summary (non-verbatim):** High-specificity phenotype flow

**Response:** The revised supplementary flow shows 453 primary AMI records, partitioned into 196 development, 79 buffer, and 178 later-group records; 138 of the 178 later AMI records met the stricter discharge-text rule, while 40 did not. This rule required an MI keyword plus an acute marker, excluded subacute and uncertain wording, and required a STEMI/NSTEMI/ST-segment subtype or explicit acute-MI phrase/site. The strict comparison used 288 CAD/angina controls, for N=426. We clarify that these were text-rule exclusions, not adjudicated misclassifications, and that discharge text is not a prospective baseline criterion.

**Changes in the revised files:** Revised manuscript pp. 5; Additional file 1 pp. 6; Results; Additional file 1, Figure S2; Additional file 2, Tables S10-S11.

## E7

**Comment summary (non-verbatim):** Fibrinogen missingness and fair same-sample comparison

**Response:** Fibrinogen was missing in 115/1,820 overall, 27/453 AMI, and 88/1,367 non-AMI CAD patients. Core-7 and Enhanced complete-case models used the same N=1,705 patients and the same validation partitions. AUCs were 0.727 and 0.737; paired  $\Delta$ AUC was 0.010 (95% CI 0.002-0.017).

**Changes in the revised files:** Revised manuscript pp. 13; Additional file 1 pp. 3; Results: Clinical incremental value and fibrinogen enhancement; Additional file 2, Table S9.

## E8

**Comment summary (non-verbatim):** Clinical baseline and incremental models

**Response:** The baseline uses age, sex, hypertension, and diabetes; smoking was unavailable. On the same fibrinogen-complete sample (N=1,705; AMI=426), the four-model results were Clinical: 0.546 (95% CI 0.515-0.574), Brier 0.187; Clinical+PIV: 0.717 (95% CI 0.689-0.745), Brier 0.166; Clinical+Core-7: 0.730 (95% CI 0.701-0.755), Brier 0.162; Clinical+Enhanced: 0.741 (95% CI 0.712-0.770), Brier 0.161. Paired  $\Delta$ AUCs were

Fibrinogen-complete cohort Clinical\_Plus\_PIV minus Clinical: 0.172 (95% CI 0.134-0.208); Fibrinogen-complete cohort Clinical\_Plus\_Core minus Clinical: 0.185 (95% CI 0.148-0.226); Fibrinogen-complete cohort Clinical\_Plus\_Enhanced minus Clinical: 0.195 (95% CI 0.157-0.231); Fibrinogen-complete cohort Clinical\_Plus\_Core minus Clinical\_Plus\_PIV: 0.013 (95% CI 0.000-0.025); Fibrinogen-complete cohort Clinical\_Plus\_Enhanced minus Clinical\_Plus\_PIV: 0.023 (95% CI 0.011-0.039); Fibrinogen-complete cohort Clinical\_Plus\_Enhanced minus Clinical\_Plus\_Core: 0.010 (95% CI 0.003-0.017). Hypertension was missing for 79/1,820 (4.34%) and diabetes for 48/1,820 (2.64%); training-fold medians were used. Full results are in Table S8.

**Changes in the revised files:** Revised manuscript pp. 13; Results: Clinical incremental value and fibrinogen enhancement; Additional file 2, Table S8.

## E9

**Comment summary (non-verbatim):** Calibration provenance and confidence intervals

**Response:** Figure 3 was regenerated from the same mean 5-fold-by-10-repeat held-out predictions used for the reported calibration statistics. The supplement now gives 10 quantile-group counts and predicted-probability summaries; Figure 5 and its source table display 95% confidence intervals consistently for all nine estimates.

**Changes in the revised files:** Revised manuscript pp. 13, 15, 16; Additional file 1 pp. 3; Methods: Internal validation and uncertainty; Figure 3; Figure 5; Additional file 2, Tables S6 and S11.

## E10

**Comment summary (non-verbatim):** Conclusion and clinical positioning

**Response:** We revised the conclusion to describe only modest internal phenotype-discrimination increments. Core-7 versus PIV was  $\Delta$ AUC 0.017; the later date-axis estimate was 0.002 with a confidence interval crossing zero. The fibrinogen complete-case increment was 0.010. We explicitly state that the findings do not establish diagnostic or triage readiness.

**Changes in the revised files:** Revised manuscript pp. 1, 2, 18; Additional file 1 pp. 4; Abstract; Discussion; Conclusions.

## R1-M1

**Comment summary (non-verbatim):** Validation rigor and temporal generalizability

**Response:** We clarified the nested repeated-CV design, training-only preprocessing/tuning, arithmetic mean of 10 held-out predictions, and fixed-prediction bootstrap scope. The date-axis estimate is now labeled exploratory and not formal temporal validation; the attenuation and timing-linkage limitation are explicit.

**Changes in the revised files:** Revised manuscript pp. 6, 7; Additional file 1 pp. 2; Methods: Internal validation and uncertainty; Temporal sensitivity analysis; Discussion.

## R1-M2

**Comment summary (non-verbatim):** Core predictor collinearity and stability

**Response:** The maximum absolute Core-7 Spearman correlation was 0.4685, between neutrophils and monocytes. In 300 development bootstrap resamples, selection frequencies were neut 100%, lymph 100%, mono 100%, plt 100%, mpv 99%, rdw 100%, hb 95%. We describe this as no strong/extreme pairwise correlation, not absence of collinearity, and retain all prespecified variables.

**Changes in the revised files:** Revised manuscript pp. 8, 15; Results: Predictor stability; Additional file 1, Figure S3; Additional file 2, Table S7.

### R1-M3

**Comment summary (non-verbatim):** Clinical covariate missingness and handling

**Response:** Age was rebuilt using the locked birth-date/reference-date rule and was available for 1,820/1,820 patients. Hypertension was missing in 79/1,820 (4.34%) and diabetes in 48/1,820 (2.64%); their medians were estimated within training folds. Smoking was unavailable and was not added.

**Changes in the revised files:** Revised manuscript pp. 9, 10, 13; Methods: Age derivation and internal validation; Table 1; Additional file 2, Table S3.

### R1-M4

**Comment summary (non-verbatim):** Traditional-index benchmarks

**Response:** The supplement reports the seven existing repeated nested cross-validation estimates with 95% CIs: NLR 0.681 (95% CI 0.652-0.709); PLR 0.618 (95% CI 0.586-0.649); MLR 0.650 (95% CI 0.621-0.681); SII 0.703 (95% CI 0.672-0.732); SIRI 0.697 (95% CI 0.669-0.726); PIV 0.708 (95% CI 0.678-0.735); HRR 0.567 (95% CI 0.537-0.597). PIV remains the prespecified primary comparator; these do not represent seven new primary tests.

**Changes in the revised files:** Revised manuscript pp. 6; Methods: Conventional indices; Additional file 2, Table S4.

### R1-M5

**Comment summary (non-verbatim):** Multiple source records and prior CAD/MI/revascularization history

**Response:** We distinguish 2,548 source records, 2,279 unique patients, 2,010 patients with one row, and 269 with two source rows. The available history fields did not have validated index-relative semantics, and no validated dedicated prior CAD, prior MI, prior PCI, or CABG field was recovered. We therefore do not invent or report these as verified prior-history covariates.

**Changes in the revised files:** Revised manuscript pp. 4; Methods: Eligibility and study population; Results: Study population; Additional file 1.

### R1-M6

**Comment summary (non-verbatim):** Fairness of the PIV comparison

**Response:** PIV was prespecified as the principal conventional comparator and used the  $\log(1+PIV)$  transformation, training-fold standardization, the same primary cohort, and the same outer partitions as Core-7. Paired comparisons use the same patient-level mean held-out predictions; the manuscript describes phenotype discrimination rather than clinical superiority.

**Changes in the revised files:** Revised manuscript pp. 6; Additional file 1 pp. 3; Methods: Conventional inflammatory indices; Internal validation and uncertainty; Results.

### R1-M7

**Comment summary (non-verbatim):** Troponin comparison

**Response:** We did not fit a troponin model or pool assay results. The recovered data lacked sufficiently complete assay-platform/manufacture details, ULN harmonization, validated serial change, and index timing; CK-MB was

present only sparsely. We acknowledge that incremental value beyond a standardized contemporary troponin pathway cannot be established.

**Changes in the revised files:** Revised manuscript pp. 16, 17; Additional file 1 pp. 1; Methods: Outcome definition; Discussion: limitations.

## **R1-m1**

**Comment summary (non-verbatim):** Consecutive screening and study period

**Response:** The source dataset was described as covering patients undergoing coronary angiography from 1 January 2020 through 1 January 2026, but the extraction query and complete eligible-patient screening ledger were not recovered. We therefore do not claim consecutive enrollment and state that the date range/eligibility could not be independently machine-verified.

**Changes in the revised files:** Revised manuscript pp. 4, 18; Methods: Study design and setting; Discussion: limitations.

## **R1-m2**

**Comment summary (non-verbatim):** CBC timing clarification

**Response:** We report the exact completeness-based source-row selection hierarchy and explicitly state that CBC-to-index-hospitalization timing could not be reconstructed uniformly. We make no claim of first admission, pre-angiography, pre-diagnostic, or pre-treatment sampling.

**Changes in the revised files:** Revised manuscript pp. 5, 9; Methods: Predictors and laboratory measurements; Figure 1; Discussion.

## **R1-m3**

**Comment summary (non-verbatim):** Informed consent for retrospective research

**Response:** The inconsistent statement that all participants provided informed consent was removed. The Declarations now state ethics approval K202602-10 and that the Ethics Committee waived informed consent for this retrospective study, with compliance with relevant guidelines and regulations.

**Changes in the revised files:** Revised manuscript pp. 19, 20; Declarations: Ethics approval and Consent to participate.

## **R2-1**

**Comment summary (non-verbatim):** Decision-curve analysis for clinical incremental models

**Response:** Exploratory DCA was added in Supplementary Figure S1 using the same N=1,705 patients (AMI=426), four clinical models, treat-all/treat-none, and thresholds 0.05-0.50. No threshold was optimized. We describe net-benefit differences as exploratory and do not claim clinical utility or treatment benefit.

**Changes in the revised files:** Revised manuscript pp. 8, 14; Additional file 1 pp. 3, 5; Results: exploratory sensitivity analyses; Additional file 1, Figure S1; Additional file 2, Table S8.

## **Ethics clarification (contextual audit item; not counted among Editor comments 1-10)**

The Ethics approval and consent declarations now give approval number K202602-10, the retrospective informed-consent waiver, and compliance with relevant guidelines and regulations. Approval documents were not included in this public-safe package.